WORKED SOLUTION

STEP III 1993, Question 7

Pure • Hyperbolic functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q7

The key idea

Expand both double-angle sinhs. The second equation then gives $\cosh x$ directly in terms of y, and dividing the first by it gives $\sinh x$ in terms of y too. The identity $\cosh^2 x - \sinh^2 x = 1$ eliminates x completely, leaving an equation in $\sinh^2 y$ alone. After that it is a matter of locating the right root of a cubic and pinning it down numerically.

Solution

Eliminating x

Using $\sinh 2u = 2\sinh u \cosh u$, the two equations become

$$2\sinh x \cosh x = \cosh y, \qquad 2\sinh y \cosh y = 2\cosh x.$$

The second gives

$$\cosh x = \sinh y \cosh y.$$

Since $\cosh x \geq 1 > 0$, we may divide the first equation by $2\cosh x$:

$$\sinh x = \frac{\cosh y}{2\sinh y \cosh y} = \frac{1}{2\sinh y},$$

where dividing by $\cosh y$ is fine because $\cosh y > 0$, and $\sinh y \neq 0$ because $\sinh y \cosh y = \cosh x \neq 0$.

Now substitute both into $\cosh^2 x - \sinh^2 x = 1$. Write $t = \sinh^2 y$, so that $\cosh^2 y = 1 + t$:

$$t(1+t) - \frac{1}{4t} = 1.$$

Multiplying by $4t$ (which is non-zero) gives $4t^2 + 4t^3 - 1 = 4t$, that is,

$$4t^3 + 4t^2 - 4t - 1 = 0,$$

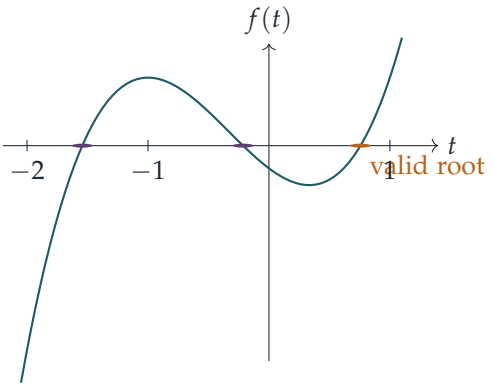
so $t = \sinh^2 y$ is a root of the given cubic, as required.

At most one valid solution

Let $f(t) = 4t^3 + 4t^2 - 4t - 1$. Since $t = \sinh^2 y$ with $\sinh y \neq 0$, a valid root must satisfy $t > 0$. Now

$$f(-2) = -9, \quad f(-1) = 3, \quad f(0) = -1, \quad f(1) = 3,$$

so by the intermediate value theorem the cubic has a root in each of $(-2, -1)$, $(-1, 0)$ and $(0, 1)$. A cubic has at most three roots, so these are all of them, and exactly one root is positive.



Ask yourself A positive t gives two values $\sinh y = \pm\sqrt{t}$. Why does only one survive?

From $\cosh x = \sinh y \cosh y$ and $\cosh x > 0$, $\cosh y > 0$, we need $\sinh y > 0$. So $\sinh y = +\sqrt{t}$, and since sinh is strictly increasing this gives exactly one y, namely $y > 0$. Hence the cubic gives at most one valid y (and then $\sinh x = 1/(2\sinh y)$ fixes x uniquely, as sinh is one-to-one).

Locating and computing t

We have

$$f(0.7) = 1.372 + 1.96 - 2.8 - 1 = -0.468 < 0,$$
$$f(0.8) = 2.048 + 2.56 - 3.2 - 1 = 0.408 > 0,$$

so by the intermediate value theorem the positive root lies between 0.7 and 0.8.

We use the Newton–Raphson iteration

$$t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)}, \qquad f'(t) = 12t^2 + 8t - 4,$$

starting from $t_0 = 0.75$, the midpoint. Here $f(0.75) = -0.0625$ and $f'(0.75) = 8.75$, so

$$\begin{aligned} t_1 &= 0.75 + \frac{0.0625}{8.75} = 0.7571429, \\ t_2 &= 0.7570685, \\ t_3 &= 0.75706846, \end{aligned}$$

and t_4 agrees with t_3 to all the figures shown. To confirm six decimal places, we check the sign change

$$f(0.7570680) < 0 < f(0.7570690),$$

so the root lies in that interval and

$$t = 0.757068 \quad \text{(6 d.p.)}.$$

(A slower but perfectly good alternative: rearrange as $t^2(4t+4) = 4t+1$, i.e. $t = \sqrt{(4t+1)/(4t+4)}$. Near the root the derivative of the right-hand side is about 0.16, so this iteration converges, gaining a little under one decimal place per step.)

Finding y and x

Taking the positive square root, $\sinh y = \sqrt{0.7570685} = 0.870097$, so

$$y = \operatorname{arsinh}(0.870097) = \ln(0.870097 + \sqrt{1.757068}) = 0.786474,$$

and

$$\sinh x = \frac{1}{2\sinh y} = 0.574649, \qquad x = \ln(0.574649 + \sqrt{1.330221}) = 0.546965.$$

Checking in the original equations:

$$\begin{aligned} \sinh 2x &= 1.325545 = \cosh y, \\ \sinh 2y &= 2.306704 = 2\cosh x, \end{aligned}$$

so both equations hold. To four decimal places,

$$x = 0.5470, \qquad y = 0.7865.$$

Checks

- The other two roots of the cubic are about -0.2140 and -1.5431: both negative, as the argument requires. The product of all three roots is $\frac{1}{4}$, which matches $0.757068 \times 0.2140 \times 1.5431 \approx 0.250$.
- A direct numerical scan over $y \in [-5, 5]$, taking $x = \pm\operatorname{arcosh}(\frac{1}{2}\sinh 2y)$ from the second equation and testing the first, finds a single solution, at $y \approx 0.7865$ with $x > 0$.
- The negative choice $\sinh y = -\sqrt{t}$ would make $\sinh y \cosh y < 0$, which can never equal $\cosh x$: a quick confirmation that it really is invalid.

Common slips

- Squaring early (for instance squaring both original equations) and then forgetting to discard the sign that doesn't work.
- Stopping the iteration as soon as two iterates agree to six places without checking the sign change across the rounding interval.
- Rounding t to six places before taking square roots and logarithms. Carry extra figures and round only the final x and y.

Beyond STEP

The two iterations here converge at very different rates. A fixed-point iteration $t \mapsto g(t)$ converges linearly, with error shrinking by about $|g'(\text{root})|$ each step, while Newton–Raphson converges quadratically, roughly doubling the number of correct digits each step. Making this precise uses the mean value theorem and Taylor's theorem, and it is the starting point of numerical analysis. For the analysis behind it, see Abbott, *Understanding Analysis*.

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